



BRAC UNIVERSITY
Department of Computer Science & Engineering
MID TERM EXAMINATION, Summer 2025
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 40 (20×2)
Time: **70 Minutes**

- There are **two(2)** questions per group. Answer any **one(1)** from **each group**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.

Name:	ID:	Section:
-------	-----	----------

Group 1

Answer **any one** from Question 1 & 2

Question 1: [CO3]

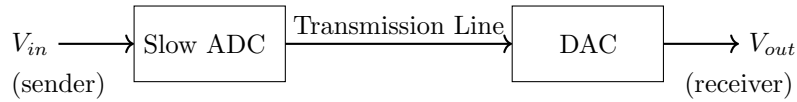


Figure 1: A lossless digital communication

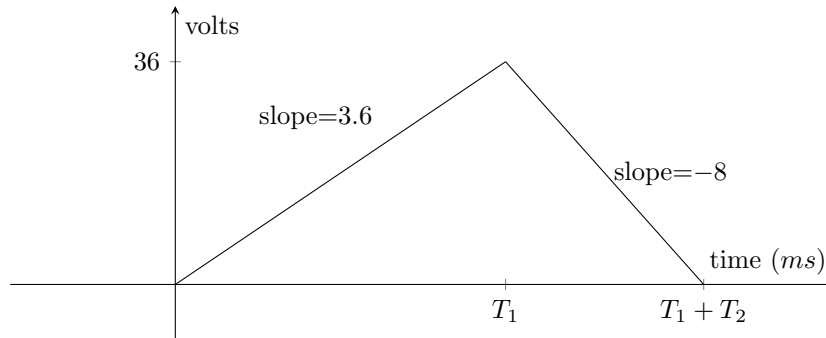


Figure 2: Characteristics of the Slow ADC

In the above communication process shown in figure 1, the **ADC** converts the V_{in} to a **4-bit** digital signal. For the characteristics graph shown in figure 2, $R = 2.5 \text{ k}\Omega$, $C = 2 \mu\text{F}$.

(a)	Find the input & reference voltage to the ADC .	[5]
(b)	Find the total conversion time & the binary output .	[6]

Now, the receiver uses a **4-bit R2R DAC** with the **same** reference voltage as found in (a). Also, $R_f = R = 1 \text{ k}\Omega$.

(c)	Find the output analog voltage (V_{out}) at the receiver for the digital input from the sender found in (b).	[3]
(d)	Determine the step size & maximum output of the DAC .	[4]
(e)	Find the percentage error between V_{in} & V_{out} .	[2]

Question 2: [CO3]

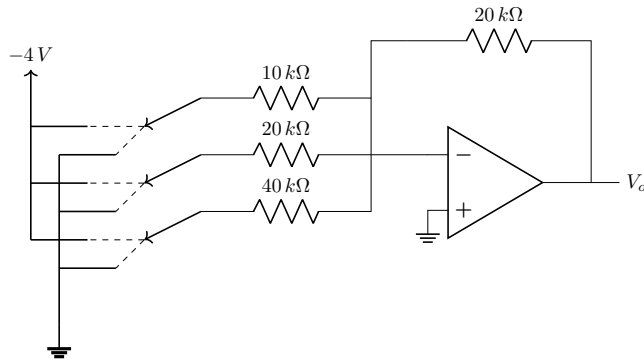


Figure 3: A 3-bit BWR DAC

The **Flash ADC** in figure 4 has a reference voltage, $V_{REF} = 32\text{ V}$. The output of the **DAC** in figure 3 is used as the input (V_{in}) of the ADC.

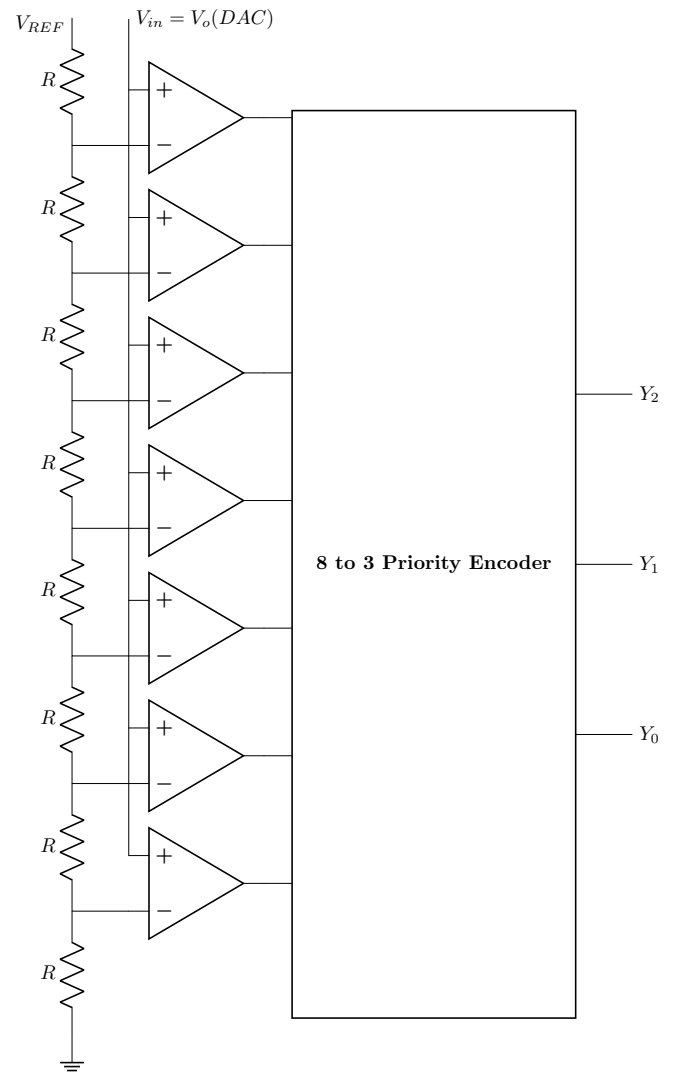


Figure 4: 3-bit Flash ADC

(a)	Determine the resolution (step size) of both the BWR DAC & Flash ADC .	[3]
(b)	(i) Suppose the Flash ADC in figure 4 has an output of '001'. Find the range of V_{in} that produces this output. (ii) Determine all possible input combinations for the DAC that will generate V_{in} (V_o of DAC) in this range.	[4+5]
(c)	What modifications can be made in the DAC circuit in figure 3, to get the same outputs from the ADC as the inputs to the DAC in (b)? [Hint: Try making changes to equate the step sizes]	[5]
(d)	If we wish to design a 4-bit Flash ADC similar to the figure, comment on the hardware changes required.	[3]

Group 2

Answer **any one** from Question 3 & 4

Question 3: [CO3]

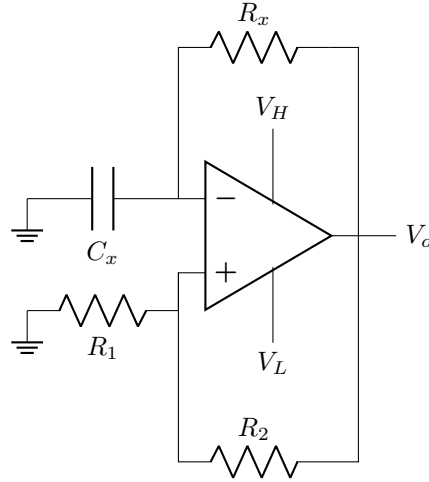


Figure 5: A Square Wave Generator

The circuit above generates a square wave with frequency **1 kHz** and **50%** duty cycle. Here, $R_1 = 3\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, $V_H = +10\text{ V}$, $V_L = -10\text{ V}$.

(a)	Find the time constant & the capacitance C_x if $R_x = 10\text{ k}\Omega$.	[4]
(b)	Find the ratio of R_1 & R_2 to double the frequency	[6]

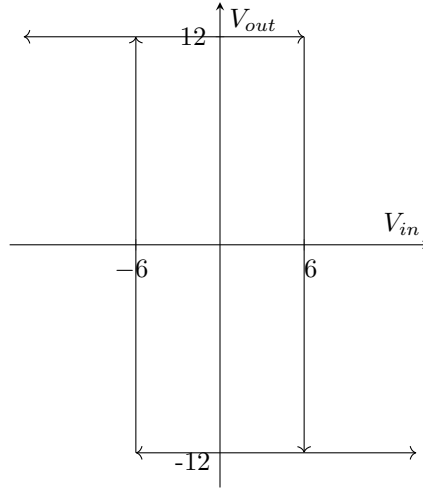


Figure 6: VTC of a Schmitt Trigger

(c)	For figure 6, find— the type of the Schmitt Trigger, hysteresis width (V_{HW}), bias voltages (V_H , V_L), thresholds (V_{TH} & V_{TL}), shift voltage (V_S).	[3]
(d)	Design a Schmitt Trigger with the same VTC, but shifted by +4 V . Draw the new VTC and the circuit . [Hint: Find the ratio of R_1 & R_2 first]	[5]
(e)	Apply the input $11\sin(2\pi ft)$ to the Schmitt Trigger & VTC designed in (d) and plot the V_{out} .	[2]

Question 4: [CO3]

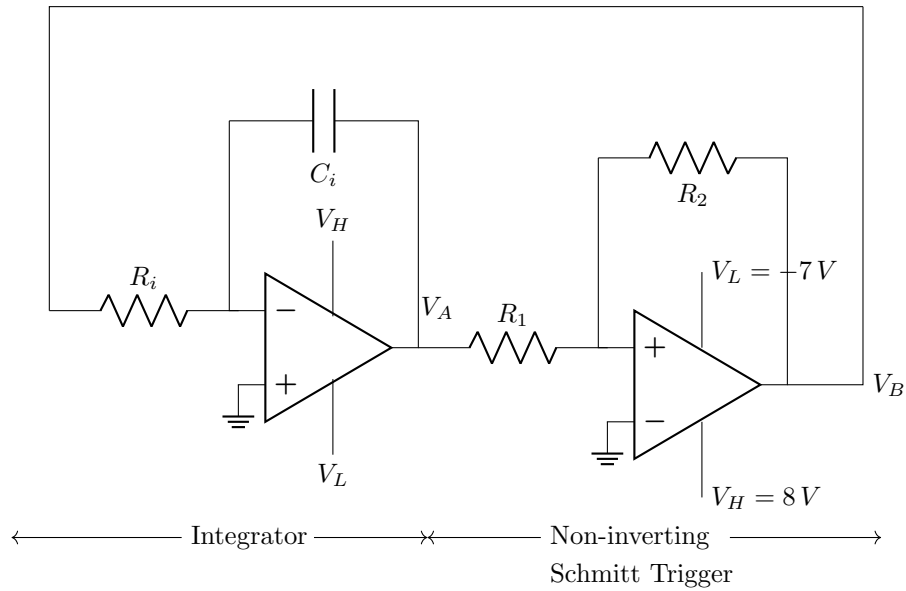


Figure 7: A Triangular Wave Generator

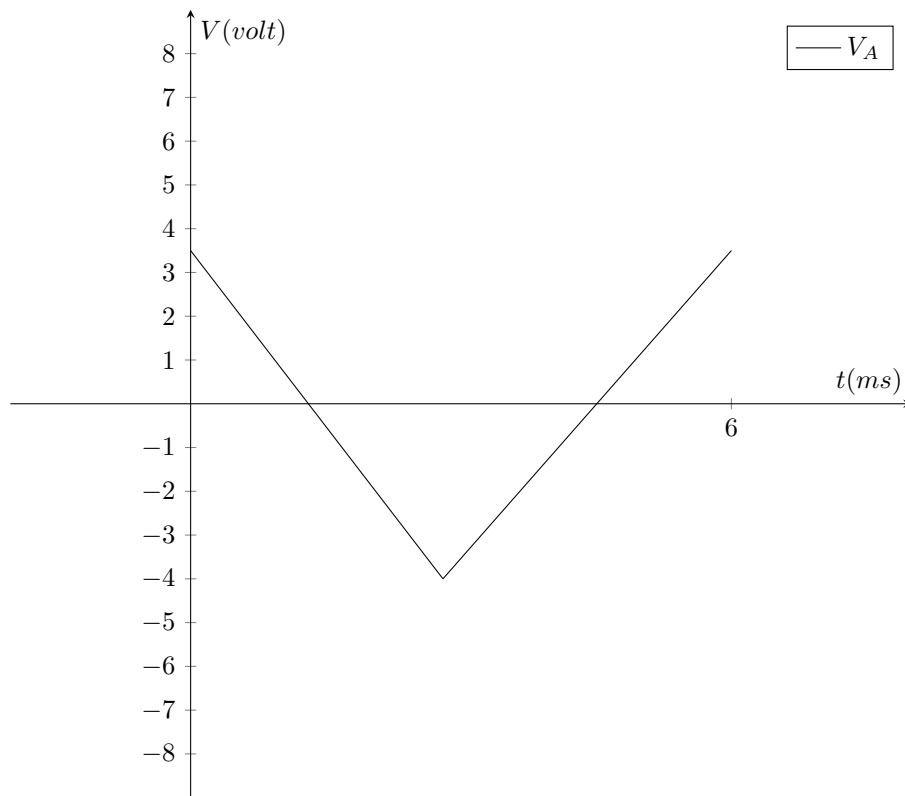


Figure 8: Output of the Triangular Wave Generator

For the triangular wave shown—

(a)	Find the time period & duty cycle . If $R_i = 1\text{ k}\Omega$, find C_i .	[8]
(b)	Draw the corresponding square wave , output from the non-inverting Schmitt Trigger of the triangular wave generator (<i>on figure 8</i>). Find its duty cycle .	[5]
(c)	Draw the VTC of the Schmitt Trigger in figure 7.	[2]
(d)	Consider a separate Schmitt Trigger has the similar VTC as in (c), but shifted by -2 V . Design the circuit and draw the output vs time graph if the input $V_i = 5\sin(\omega t)$ was applied to it.	[5]